

Customer Churn Prediction System

DOCUMENTATION

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1. Introduction

1.1 Purpose

This system predicts customer churn in telecom companies using machine learning. It employs a multi-agent architecture where specialized agents handle different tasks.

1.2 Scope

- Predicts if a customer will leave (churn)
- Provides risk level assessment (Low/Medium/High)
- Generates retention recommendations
- Supports bulk analysis via CSV upload

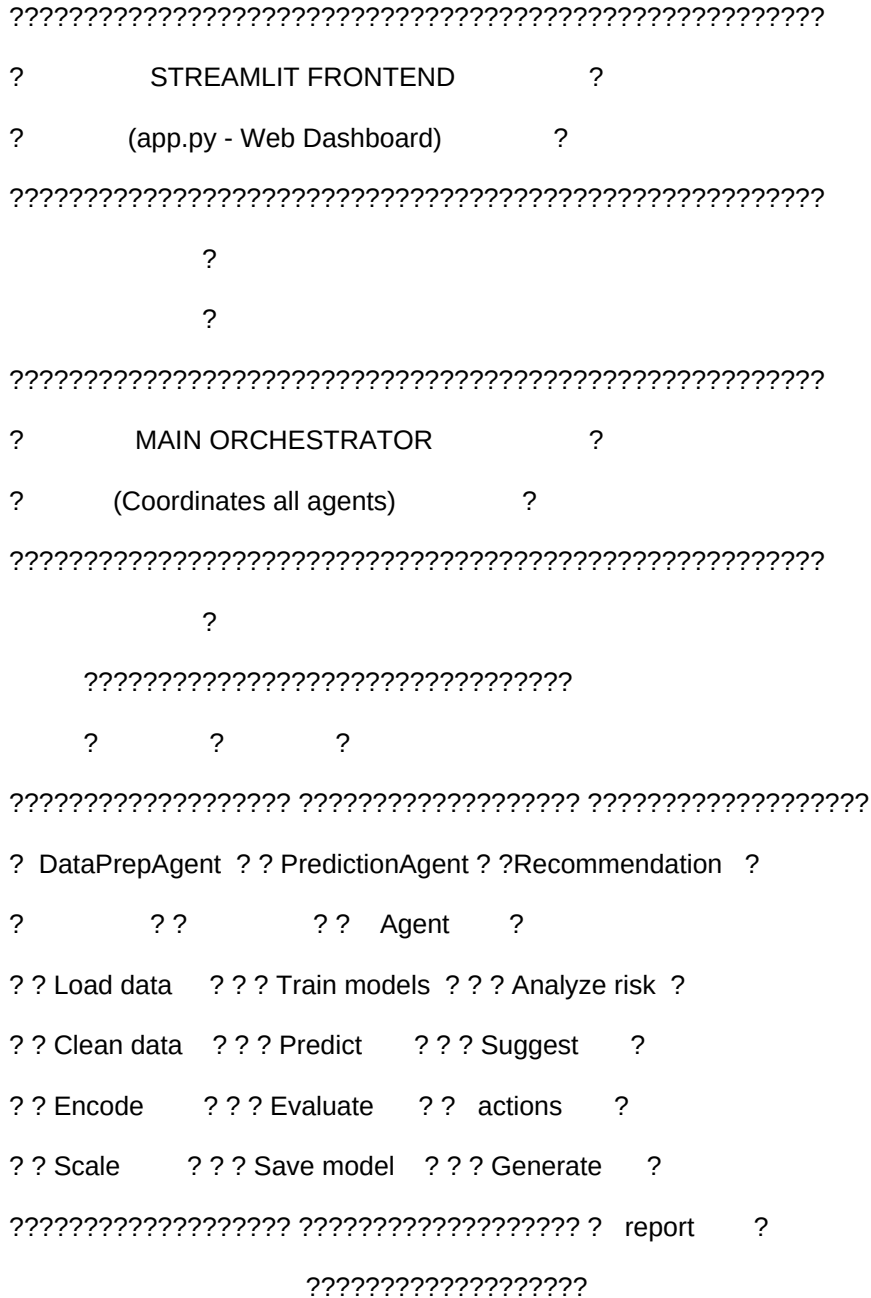
1.3 Target Users

- Telecom companies
 - Customer retention teams
 - Data analysts
 - Business managers
-

2. System Architecture

Customer Churn Prediction System

2.1 High-Level Architecture



2.2 Data Flow

Input Data

?

?

[DataPrepAgent]

?

?? Remove duplicates

?? Handle missing values

Customer Churn Prediction System

?? Encode categorical features

?? Scale numerical features

?

?

[PredictionAgent]

?

?? Split train/test

?? Train 3 models

?? Evaluate metrics

?? Select best model

?

?

[RecommendationAgent]

?

?? Calculate risk level

?? Generate recommendations

?? Create report

?

?

Output (Dashboard/CSV)

3. Installation

3.1 Prerequisites

- Python 3.8 or higher
- pip package manager
- Git (optional)

3.2 Install Dependencies

Navigate to project directory

```
cd "C:\Users\Lenovo\Desktop\Yuktamedia\ML project\1st"
```

Install all requirements

Customer Churn Prediction System

pip install -r requirements.txt

3.3 Required Packages

Package	Version	Purpose
pandas	>=1.5.0	Data manipulation
numpy	>=1.22.0	Numerical operations
scikit-learn	>=1.1.0	Machine learning
xgboost	>=1.6.0	Gradient boosting
streamlit	>=1.28.0	Web dashboard
plotly	>=5.15.0	Interactive charts
joblib	>=1.1.0	Model persistence

4. Project Structure

```
1st/
?
??? agents/                # AI Agents package
?  ??? __init__.py         # Package exports
?  ??? base_agent.py       # Abstract base class
?  ??? data_prep_agent.py  # Data preparation agent
?  ??? prediction_agent.py # ML prediction agent
?  ??? recommendation_agent.py # Recommendation agent
?
??? models/                # ML Models package
?  ??? __init__.py         # Package exports
?  ??? churn_model.py      # Model management
?
??? data/                  # Dataset directory
?  ??? telco_churn.csv      # Main dataset (5,043 records)
?  ??? test_upload.csv      # Test CSV for upload
?  ??? sample_customers_1.csv # Sample data 1
?  ??? sample_customers_2.csv # Sample data 2
?  ??? sample_customers_3.csv # Sample data 3
?
```

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```
??? utils/                # Utilities package
?  ??? __init__.py         # Package exports
?  ??? helpers.py          # Helper functions
?
??? app.py                 # Streamlit web application
??? main.py                # Backend runner (CLI)
??? requirements.txt        # Python dependencies
??? SYNOPSIS.md            # Project synopsis
??? DOCUMENTATION.md       # This file
```

5. Agent Documentation

5.1 BaseAgent (Abstract Class)

File: `agents/base\_agent.py`

```
class BaseAgent(ABC):
    def __init__(self, name: str)
    def initialize(self) -> None    # Abstract
    def execute(self, data: Dict) -> Dict # Abstract
    def log(self, message: str) -> None
```

5.2 DataPrepAgent

File: `agents/data\_prep\_agent.py`

Purpose: Handles all data preprocessing tasks.

Methods:

Method	Description
`initialize()`	Setup agent resources
`execute(data)`	Run full preprocessing pipeline
`_clean_data(df)`	Remove duplicates, handle missing values
`_encode_features(df)`	Convert categorical to numerical
`_scale_features(df)`	Normalize numerical features
`prepare_single_customer(data)`	Prepare single customer for prediction

Input Format:

```
{
    'dataframe': pd.DataFrame, # Raw data
```

Customer Churn Prediction System

```
'fit_scaler': bool      # True for training, False for prediction
}
```

Output Format:

```
{
  'dataframe': pd.DataFrame,    # Processed data
  'feature_columns': list,      # Feature names
  'label_encoders': dict,       # Encoders for categories
  'scaler': StandardScaler      # Scaler for numbers
}
```

5.3 PredictionAgent

File: `agents/prediction\_agent.py`

Purpose: Trains ML models and makes predictions.

Methods:

Method	Description
`initialize()`	Setup 3 ML models
`execute(data)`	Train and evaluate models
`predict(X)`	Make predictions on new data
`save_model(filepath)`	Save best model to disk
`load_model(filepath)`	Load model from disk

Models Used:

1. LogisticRegression(max_iter=1000)
2. RandomForestClassifier(n_estimators=100)
3. XGBClassifier(n_estimators=100, learning_rate=0.1)

Metrics Calculated:

- Accuracy
- Precision
- Recall
- F1 Score

5.4 RecommendationAgent

File: `agents/recommendation\_agent.py`

Purpose: Generates customer retention recommendations.

Methods:

Method	Description
--------	-------------

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<code>`initialize()`</code>	Setup recommendation engine
<code>`execute(data)`</code>	Generate recommendations
<code>`_get_risk_level(prob)`</code>	Determine risk level
<code>`_get_customer_recommendations(...)`</code>	Get specific actions
<code>`_generate_report(...)`</code>	Create full report

Risk Levels:

Probability	Level	Action
≥ 0.7	HIGH	Urgent intervention
≥ 0.4	MEDIUM	Offer promotions
< 0.4	LOW	Maintain service

6. API Reference

6.1 DataPrepAgent

```
from agents import DataPrepAgent
```

```
agent = DataPrepAgent()
```

```
agent.initialize()
```

```
result = agent.execute({
```

```
    'dataframe': df,
```

```
    'fit_scaler': True
```

```
})
```

6.2 PredictionAgent

```
from agents import PredictionAgent
```

```
agent = PredictionAgent()
```

```
agent.initialize()
```

```
result = agent.execute({'dataframe': clean_df})
```

```
predictions, probabilities = agent.predict(test_df)
```

6.3 RecommendationAgent

```
from agents import RecommendationAgent
```

```
agent = RecommendationAgent()
```

```
agent.initialize()
```

```
result = agent.execute({
```

```
    'predictions': predictions,
```

Customer Churn Prediction System

```
'probabilities': probabilities,  
'feature_columns': columns,  
'customer_data': test_df,  
'metrics': metrics,  
'best_model_name': 'logistic_regression'  
})
```

7. Usage Guide

7.1 Run Web Dashboard

```
cd "C:\Users\Lenovo\Desktop\Yuktamedia\ML project\1st"  
  
streamlit run app.py  
  
Open browser: `http://localhost:8501`
```

7.2 Run Backend Only

```
cd "C:\Users\Lenovo\Desktop\Yuktamedia\ML project\1st"  
  
python main.py
```

7.3 Dashboard Tabs

Tab	Function
Predict	Single customer prediction
Analytics	Data visualization
Models	Model comparison
Bulk Analysis	CSV upload & analysis

7.4 Upload CSV Format

Required columns:

customerID,gender,SeniorCitizen,Partner,Dependents,tenure,
PhoneService,MultipleLines,InternetService,OnlineSecurity,
OnlineBackup,DeviceProtection,TechSupport,StreamingTV,
StreamingMovies,Contract,PaperlessBilling,PaymentMethod,
MonthlyCharges>TotalCharges,Churn

8. Dataset

Customer Churn Prediction System

8.1 Source

Telco Customer Churn Dataset (IBM Sample Data)

8.2 Size

- Records: 5,043 customers
- Features: 20 columns
- Target: Churn (Yes/No)

8.3 Feature Descriptions

Feature	Type	Description
gender	Categorical	Male/Female
SeniorCitizen	Binary	0/1
Partner	Binary	Yes/No
Dependents	Binary	Yes/No
tenure	Numerical	Months as customer
PhoneService	Binary	Yes/No
MultipleLines	Categorical	Yes/No/No phone service
InternetService	Categorical	DSL/Fiber optic/No
OnlineSecurity	Categorical	Yes/No/No internet service
OnlineBackup	Categorical	Yes/No/No internet service
DeviceProtection	Categorical	Yes/No/No internet service
TechSupport	Categorical	Yes/No/No internet service
StreamingTV	Categorical	Yes/No/No internet service
StreamingMovies	Categorical	Yes/No/No internet service
Contract	Categorical	Month-to-month/One year/Two year
PaperlessBilling	Binary	Yes/No
PaymentMethod	Categorical	Electronic check/Mailed check/Bank transfer/Credit card
MonthlyCharges	Numerical	Monthly bill amount
TotalCharges	Numerical	Total amount paid
Churn	Binary	Yes/No (target)

9. Model Details

9.1 Logistic Regression

Metric	Value
Accuracy	79.48%
Precision	62.40%

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Recall	56.55%
F1 Score	59.33%

9.2 Random Forest

Metric	Value
Accuracy	78.69%
Precision	61.93%
Recall	50.56%
F1 Score	55.67%

9.3 XGBoost

Metric	Value
Accuracy	78.20%
Precision	60.09%
Recall	52.43%
F1 Score	56.00%

10. Troubleshooting

10.1 Common Errors

Error	Cause	Solution
`ModuleNotFoundError`	Missing package	Run `pip install -r requirements.txt`
`FileNotFoundError`	Wrong path	Check `data/telco_churn.csv` exists
`ValueError: Feature names mismatch`	Churn column in upload	Remove Churn column from CSV
`ConvergenceWarning`	Model didn't converge	Increase max_iter or scale data

10.2 Port Already in Use

Find process using port 8501

```
netstat -ano | findstr :8501
```

Kill the process

```
taskkill /PID <PID> /F
```

10.3 Reset Streamlit Cache

```
streamlit run app.py --server.headless true
```

Customer Churn Prediction System

Contact

Project Location:

C:\Users\Lenovo\Desktop\Yuktamedia\ML project\1st

Run Commands:

Web Dashboard

streamlit run app.py

Backend Only

python main.py